

An exceptional theoretician, Archimedes was one of the greatest mathematicians of the ancient world. Applying his genius to solve practical problems, his original ideas gave rise to inventions, as well as writings, in geometry, mathematics, mechanics, and physics that are still valuable today.

#### MILESTONES

##### STUDENT WORKS

In Alexandria, c.250 BCE, writes the works *On the Measurement of a Circle* and *On Floating Bodies*.

##### BUILDS SHIP

In c.240 BCE, designs *Syracusia*, a giant ship built as a gift for Ptolemy III of Egypt. It sails once.

##### NUMBER SYSTEM

Devises a way of writing big numbers and calculates the number of sand grains in the Universe.

##### EUREKA MOMENT

Taking a bath, he finds how to measure the volume of irregular objects, and "Archimedes' principle."

Born in Syracuse, a Greek city-state in modern-day Sicily, Archimedes followed his father Phidias by studying astronomy and mathematics. Early in his career, he went to Egypt to study in Alexandria, the center of learning in the ancient world. Here, he may have known such brilliant minds as Eratosthenes of Cyrene (see p.21), to whom he addressed his *Method of Mechanical Theorems*. Archimedes then returned to Syracuse, where King Hieron II—who may have been a relative—became his patron.

#### Geometry genius

Archimedes was a polymath who turned his intellect to a host of problems, but his pet subject was geometry. Such was his devotion to the study of shapes that he saw his inventions as trivial in comparison. He was the first to calculate the approximate value of  $\pi$  (pi)—the ratio of a circle's circumference to its diameter—and in *On Conoids and Spheroids*, he set out how to calculate the volume of geometrical solids. Archimedes considered one of these mathematical proofs—that the volume of a

**Archimedes studied** solids, including how they behave in water. The Archimedes screw—a hand-driven spiral that drew water up to higher levels—took his name.

